

Trustworthy AI-Assisted Coral Reef Health Assessment and Temporal Change Detection Using Underwater ROV Imagery: A Lakshadweep Case Study

BY

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Introduction

Coral reefs are biodiversity rich ecosystems that are increasingly affected by bleaching, disease and other environmental stresses. Monitoring reef condition through underwater surveys and image analysis is often labour intensive. AI and computer vision provide an opportunity to automate the analysis of underwater imagery and derive quantitative reef health indicators. This project investigates a trustworthy AI assisted framework for coral reef health assessment and temporal change detection using ROV imagery from Lakshadweep

Literature Review

Paper	What they have done	Inspiration for Our Work
Monitoring of Coral Reefs Using Artificial Intelligence: A Feasible and Cost-Effective Approach	Used deep learning for automated benthic image annotation and coral monitoring.	Automated analysis of underwater imagery
Quantifying the Loss of Coral from a Bleaching Event Using Underwater Photogrammetry and AI-Assisted Image Segmentation	Combined underwater photogrammetry with AI-assisted segmentation to quantify coral loss after bleaching.	Quantitative coral condition/change estimation
A Generalized Machine Learning Model for Long-Term Coral Reef Monitoring in the Red Sea	Used ML with satellite imagery to detect long-term coral reef habitat changes.	Temporal change analysis
The Coralscapes Dataset: Semantic Scene Understanding in Coral Reefs	Developed Coralscapes, a semantic segmentation dataset with multiple benthic classes.	Coral/benthic semantic segmentation

Literature Review

Paper	What they have done	Inspiration for Our Work
Deep Learning for Multi-label Classification of Coral Conditions in the Indo-Pacific via Underwater Photogrammetry	Applied deep learning for multi-label classification of coral conditions such as healthy, compromised, dead and rubble	Multi class coral condition assessment
Multi-label Classification for Multi-temporal, Multi-spatial Coral Reef Condition Monitoring Using Vision Foundation Model with Adapter Learning	Used DINOv2 + LoRA for coral-condition classification across different spatial and temporal observations	Foundation models & transfer learning
Deep Learning for Automated Coral Reef Monitoring: A Novel System Based on YOLOv8 Detection and DeepSORT Tracking	Integrated YOLOv8 + DeepSORT to detect and track coral colonies in underwater video and compare population-level changes across years	Video detection, tracking & temporal monitoring

Literature Review

Paper	What they have done	Inspiration for Our Work
Advancing Coral Reef Monitoring: A Deep Learning Perspective on Automated Segmentation and Classification	Examined deep learning approaches for automated coral segmentation and classification and their challenges in underwater environments.	Robust segmentation & classification
Coral Reef Health Detection using ResNeSt Algorithm with Cross-Spatial Attention	Used ResNeSt with Cross-Spatial Attention for coral health/bleaching detection	Attention based coral health assessment
A Comprehensive Literature Review on Coral Reef Health Monitoring: Techniques, Status, and Emerging Trends with Special Focus on South India	Reviewed conventional, remote-sensing and AI-based coral monitoring approaches on South India and Lakshadweep	Regional context & integrated monitoring

Motivation

Integrated Coral Reef Health Assessment

- Existing AI approaches largely address detection, segmentation, condition classification, or temporal monitoring as individual tasks.
- There is a need for an integrated framework that converts underwater ROV imagery into quantitative reef-health indicators and temporal change insights

Our Plan

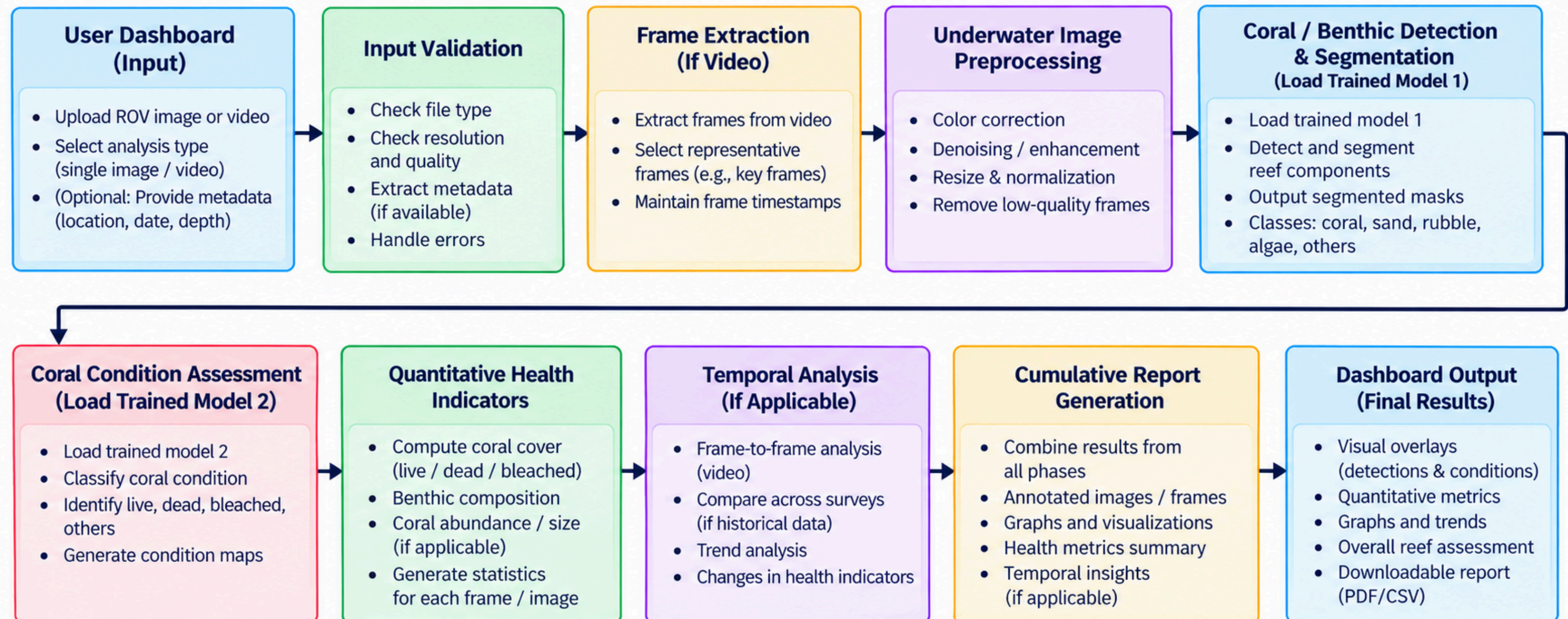
- Develop an integrated and trustworthy AI assisted framework that connects underwater coral/benthic analysis with coral condition assessment, quantitative reef-health indicators and temporal change analysis.

Problem Statement

To develop an AI-assisted framework for automated assessment of coral reef health and temporal changes using underwater ROV imagery

- Automatically identify and segment coral and other relevant benthic components from ROV imagery
- Assess coral condition by identifying indicators such as live, dead and bleached coral, based on the available dataset labels
- Estimate quantitative reef-health indicators, such as coral cover and the proportion of different coral conditions.
- Compare observations across locations and survey years to identify changes in reef condition, where comparable temporal data are available.

Proposed Architecture



Coral Datasets Used for AI-Based Reef Health Assessment

Dataset	Source	What it contains	Use in Project
Agatti + Kadmat + Kavaratti Coral Dataset	Dryad	long-term coral observations including coral genera, lifeforms, percentage coral cover, survey year, depth, transects, quadrats and photographic records collected across different sites and years.	Historical coral composition, coral-cover trends and temporal analysis
Moorea Labeled Corals	MCR-LTER	underwater reef images with expert point-level annotations identifying different coral genera, algae, sand and other reef components .	Coral image classification, pretraining and transfer learning
Moorea Orthophoto Dataset (MOD)	GitHub	underwater orthophoto image patches with annotations for coral health conditions and coral genera , along with associated depth information.	Coral health classification and image-based analysis
Coralscapes	Zenodo	Contains high-resolution underwater reef images with detailed polygon annotations representing coral, algae, sand, rubble and other benthic components.	Semantic segmentation and model pretraining
Selected CoralNet Datasets	CoralNet	underwater coral imagery accompanied by expert annotations identifying different coral, algae and substrate categories .	Additional training and transfer learning
NOAA Coral Reef Watch – Thermal Data	NOAA Coral Reef Watch	Contains SST, thermal anomalies, HotSpot, DHW and bleaching-alert information from 1985 to the present, providing a long-term record of coral thermal stress	Environmental context for coral stress and bleaching analysis

Experimental Setup

Component	Planned Setup
Dataset	Underwater coral reef ROV images/videos + available annotations & metadata
Data Preparation	Frame extraction, annotation processing, train/validation/test split
Preprocessing	Resize, normalization, enhancement & augmentation
Model	YOLO26 for coral/benthic detection & segmentation
Evaluation	Precision, Recall, mAP, IoU, F1, Inference performance
System Evaluation	Condition assessment, health indicators, temporal analysis & trustworthy-AI outputs

Conclusion

- So far, we have collected the Agatti, Kadmat and Kavaratti coral datasets, reef-fish data, and NOAA thermal-stress data, providing the main data foundation for the project.
- The reference papers have guided our methodology toward AI-based coral detection/segmentation and health classification, underwater photogrammetry and temporal change detection, and uncertainty-aware/trustworthy AI.

Next Phase: Inspiring from approach in [1] as baseline, going to implement YOLO26 directly for coral detection/segmentation, evaluate its performance on underwater coral reef imagery, and use the results to develop the subsequent coral condition assessment, quantitative health indicators, temporal change analysis, and dashboard based cumulative reporting pipeline.

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